

such as the exact location of the object, the time when it entered or exited the geofence, the time it spent at a particular position, and average time spent in travelling from and to a set position. It allows the user to set up a geographical limit as well as a time limit and a call for alert in case of any violation of rules. The zones and times applied can be varied for different objects and different days of the week, if required. The alert type can also be customised.

Geofencing software can be used by different people in a multitude of ways. For example, for a housing maintenance manager, it is important to know if all his vehicles are available on call inside boundaries of the city. Hence he can specify the boundary of the city as a mandatory zone between certain hours from Monday to Friday, or as required.

For insurance purposes, a transport manager needs to be sure that his trucks are parked in the depot overnight, not outside drivers' homes. Therefore he can specify the location on the map and make it a mandatory zone for a specific time period. Breach of this zone would trigger an alert to the concerned official, preventing misuse of time and fuel.

## Geofencing in marketing

Geofencing helps to create a communicative interaction between a brand and its customers. It is useful for businesses to promote their brands, products or offers to enlarge the market size.

One of the most common ways is to send special offers to customers when they walk or drive near a business location. For example, if you run a pizza shop and want to promote a new variety of pizza, you send out a geofence message inviting people to come in and taste the new recipe with a bottle of Coke. Or if you run a gym and want to launch a new health event or programme such as "Burn your belly fat," geofence messages can be



Fig. 2: A geofencing-based cattle finder application  
(Source: [www.cattle-watch.com](http://www.cattle-watch.com))



Fig. 3: A virtual boundary superimposed on Google Earth map  
(Source: [www.aisc.aero](http://www.aisc.aero))

used to alert users and invite them to join the programme.

Thus geofence makes push notifications popular by making the mobile messaging timely, relevant and inordinate for customer engagement.

Geofence technology provides precision customer targeting with genuine, personalised messages. It provides multi-channel businesses an opportunity to interact with nearby customers through mobile devices, split-test customer engagement at every level and gain a deeper understanding of customers' desires. It is a powerful tool to connect with new customers looking for your product as well as existing customers present in

the area. Thus geofencing enables brick-and-mortar expansion of businesses.

Imagine you enter an area covered by geofence and are running a geofence app on your mobile device. You are looking for a good pair of wireless headphones and hence key-in some relevant search words such as wireless headphones. Within seconds you receive a message or notification and ads for in-store purchase of wireless headphones. You like the headphones, find the offer attractive and walk into the store to purchase the same.

The fenced area could be as big as one km radius or just the walking distance in a mall. In rural areas, geofence is likely to cover a much wider area, given the lower population density.

The fence coordinates are hard-coded into the app. The app users get trigger notifications as they enter or exit the geofence area. The apps use various technologies to spot the change of location, most of which are generally Wi-Fi based.

Geofences are created using mapping software, which allow drawing of the geofence over the desired area using latitude and longitude coordinates or, in case of a circular geofence, a single point that forms the centre. The data can be obtained via a wireless connection, RFID and even GPS.

A geofence has a fixed location and the object using GPS (a vehicle, equipment or person) is generally moving or is movable.

For a GPS-tracked object to work with a geofence, the following conditions need to be met:

1. Creation of a geofence using mapping/fleet management software
2. Ability of the GPS-tracked object to report its position (latitude/longitude) to the used software
3. Association of the tracked object with the mapping software
4. Establishment of a rule that triggers an alert when the tracked object enters (or leaves) the geofenced area